

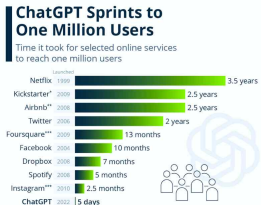
LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models

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MicrosoftCorporation

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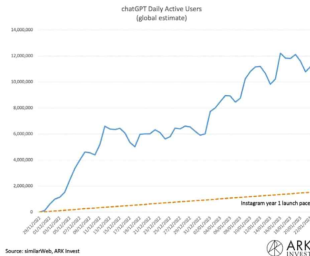
ChatGPT's infleunce



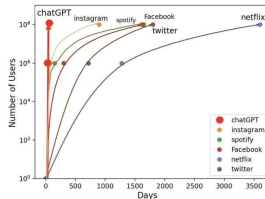
* one million backers ** one million nights booked *** one million downloads
Source: Company announcements via Business Insider/LinkedIn



statista

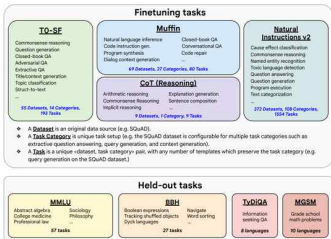


ChatGPT's influence



ChatGPT's influence

- Powerful generalization and reasoning capabilities of LLMs.
- Suitable prompts is crucial, they involves techniques such as chain-of-thought, in-context learning, and retrieving related documents (Wei et al., 2022; Chase, 2022).



```
def eat_at_restaurant():
    """Made ordered one chicken meal that costs $12, 5
    packs of milk that costs $3 each, 4 apples that cost
    $1.50 each and 3 frozen pizzas. Made paid a total of
    $50. How many boxes of pizza did Mark order if each
    box costs $8.50?"""

    total_paid = 50
    cost_of_pizza_per_box = 8.50
    cost_of_meal = 12
    cost_of_milk_per_pack = 3
    cost_of_apples_per_apple = 1.50
    packages_of_milk = 5

    return (total_paid - cost_of_meal - cost_of_milk_per_pack *
            packages_of_milk - cost_of_apples_per_apple * 4) / cost_of_pizza_per_box


# Translating from C to Python:
long long int sum_pairwise_products (int a []) {
    long long int sum = 0;
    for (int i = 1; i < a.length; i++)
        sum = sum + a[i-1] * a[i];
    return sum;
}

def sum_pairwise_products(x):
    sum = 0
    for i in range(1, len(x)):
        sum = sum + x[i-1] * x[i]
    return sum
```

```

prompt
# Given a string representing musical notes in a
# special ASCII format.
# parse this string and compute a total length of
# the music in beats.
# Legend
# 'O' - whole note, lasts four beats
# 'H' - half note, lasts two beats
# 'Q' - quarter note, lasts one beat
notes = "O O O O O O O O O O"

# 1. Split the string into a list of notes.
# 2. Iterate over the list and compute the total
# length of the music.
# 3. Print the result!

notes = notes.split()

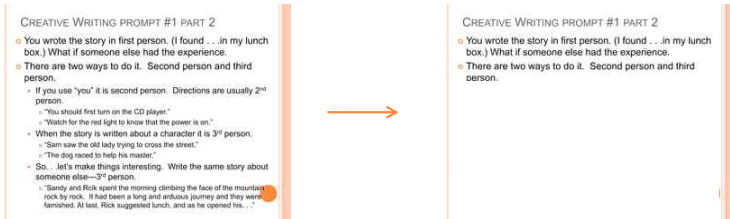
# 1
total_length = 0
for notes in notes:
    if notes == "O":
        total_length += 4
    if notes == "H":
        total_length += 2
    if notes == "Q":
        total_length += 1

print(total_length)

```

Background

- However, activating LLM's domain specific knowledge, they require long prompts (i.e. in-context learning, chain-of-thought).
- Approaches that attempt to reduce the length of original prompts have emerged lately while preserving essential information.



Background

- Removing tokens with lower perplexity has a relatively minor impact on the LLM's comprehension of the context.
- Motivated by this, Li(2023) propose Selectoive-Context.
 - compute self-information of lexical unit (sentences, phrases, or tokens) and drop the less informative content

Context: Large Languages Models (LLMs) *{trained-on massive-corpora-of-texts}* have shown their ability to perform new tasks from textual instructions or from a few examples, which resulting in a line of work that focuses on further scaling these models. These efforts are based on the assumption *{that more parameters will lead to better performance.}*

Query: What's the assumption behind?

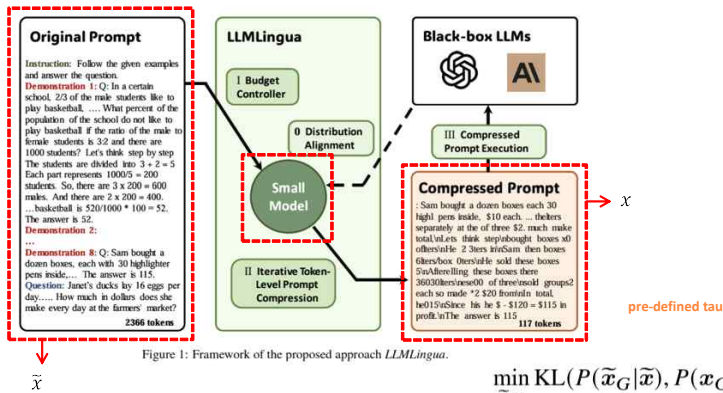
GPT: Further scaling Large Language Models will lead to better performance on a wide range of tasks.

self-information

$$I(x_i) = -\log_2 P(x_i|x_0, x_1, \dots, x_{i-1})$$

Limit : ignore the independence between correspondence contents
& neglect correspondence between the LLM and the small language model

Problem Formulation



$$\min_{\tilde{x}, \tau} \text{KL}(P(\tilde{x}_G|\tilde{x}), P(x_G|x)),$$

LLMLingua

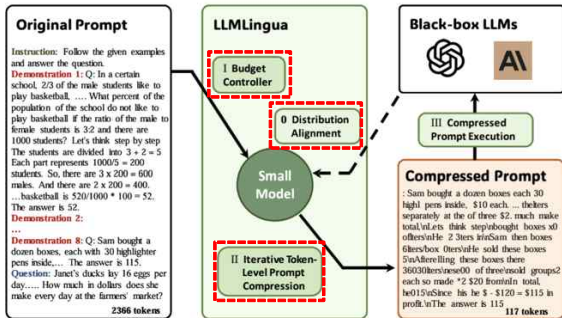


Figure 1: Framework of the proposed approach LLMLingua.

0. Distribution Alignment

- To narrow the gap between the distribution LLM and that of small LM.
- Aligning the two distribution via instruction tuning
- Pre-trained small language model M_s
- A pair of instruction x_i and LLM generated text $y_{i,LLM}$

$$\min_{\theta_s} \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \mathcal{L} (x_i, y_{i,LLM}; \theta_{M_s}) \right]$$

Primary knowledge

- PPL (Perplexity)
 - One of the most common metric for evaluating language models.
 - Perplexity is defined as the exponentiated average negative log-likelihood of a sequence.
 - In terms of information entropy, tokens with lower PPL contribute less to the entropy gains of the language model.

Hugging Face is a startup based in New York City and Paris

$p(\text{word}|\text{context})$

$$\text{PPL}(X) = \exp \left\{ -\frac{1}{t} \sum_i^t \log p_{\theta}(x_i | x_{<i}) \right\}$$

X : a tokenized sequence

I. Budget Controller

- Allocate different budgets
(i.e. instructions, demonstrations, and questions)
- Two considerations
 - Instruction and question - direct influence on the generated results , less redundant.
 - > Smaller compression ratios
 - High compression ratio - sentence-level dropout with preserving a linguistic integrity.
 - > Multiple demonstration : demonstration-level control

I. Budget Controller

Algorithm 1 Pseudo code of Budget Controller.

Input: A small language model M_s ; the original prompt $x = (x^{\text{ins}}, x^{\text{dems}}, x^{\text{que}})$.

- 1: Set the selected demonstration set $\mathcal{D} = \phi$.
- 2: Get demonstration compression rate τ_{dems} by Eq.(2)
- 3: Calculate the perplexity of each demonstration via M_s .
- 4: Rank all demonstrations in descending order of their perplexity as a list $(x_{(1)}^{\text{dem}}, \dots, x_{(N)}^{\text{dem}})$, where N is the number of demonstrations, $x_{(i)}^{\text{dem}}$ is the i -th demonstration.
- 5: **for** $i = 1$ **do**
- 6: **if** $\tilde{L}_{\mathcal{D}} > k \cdot \tau_{\text{dems}} L_{\text{dems}}$ **then**
- 7: Break.
- 8: **end if**
- 9: Append $x_{(i)}^{\text{dem}}$ to $\mathcal{D}$.
- 10: $i = i + 1$
- 11: **end for**
- 12: Allocate remaining budget to x^{ins} and x^{que} via Eq.(3)

Output: The subset of demonstrations $\mathcal{D}$ obtained from coarse-grained compression; Additional budget $\Delta\tau_{\text{ins,que}}$ for the instruction and the question.

$$\tau_{\text{dems}} = \frac{\text{total length } \tau L - (\text{pre-defined tau } (\tau_{\text{ins}} L_{\text{ins}} + \tau_{\text{que}} L_{\text{que}}))}{L_{\text{dems}}}$$

$$0.696 = \frac{0.7 \times 1855 - (0.85 \times 30 + 0.9 \times 25)}{1800}$$

$$\Delta\tau = \frac{\text{control coefficient } k \cdot \tau_{\text{dems}} L_{\text{dems}} - \tilde{L}_{\mathcal{D}}}{L_{\text{ins}} + L_{\text{que}}}$$

$$\text{coressponding compression ratio} = \frac{k \cdot 1252.8 - 1260}{55}$$

II. Iterative Token-level Prompt Compression

- Perplexity is based on the independence assumption.
- Original prompt after demonstration-level compression : $x' \longrightarrow S = \{s_1, s_2, \dots, s_m\}$
- Each segment is concatenated to the subsequent segment, conditional probabilities for each segment $p(\tilde{s}_j)$ are obtained.

$$p(\tilde{x}) = \prod_{i=1}^{\tilde{L}} p(\tilde{x}_i | \tilde{x}_{<i})$$

$$\approx p(x') = \prod_{i=1}^{L'} p(x_i | \tilde{x}_{<i}, \bar{x}_{<i}),$$



$$p(\tilde{s}_j) = \prod_{i=1}^{\sum_k^j \tilde{L}_{s,k}} p(\tilde{s}_{j,i} | \tilde{s}_{j,<i}, \tilde{s}_{<j})$$

$$\approx \prod_{i=1}^{L_{s,j} + \sum_k^{j-1} \tilde{L}_{s,k}} \boxed{p(s_{j,i} | s_{j,<i}, \tilde{s}_{<j})},$$

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- Original prompt after demonstration-level compression : $x' \longrightarrow S = \{s_1, s_2, \dots, s_m\}$
- Each segment is concatenated to the subsequent segment, conditional probabilities for each segment $p(\tilde{s}_j)$ are obtained.
- The compression ratio threshold γ_j are dynamically calculated based on the PPL
- tokens in each s_j with the PPL greater than γ_j are retained in compressed prompt.

$$\tau_{s_j} = \begin{cases} \tau_{\text{ins}} + \Delta\tau, & \text{if } s_j \text{ from } x^{\text{ins}}, \\ \tau_{\text{dems}}, & \text{if } s_j \text{ from } x^{\mathcal{D}}, \\ \tau_{\text{que}} + \Delta\tau, & \text{if } s_j \text{ from } x^{\text{que}}. \end{cases}$$

coressponding
compression ratio

$$\tilde{s}_j = \{s_{j,i} | p(s_{j,i}) > \gamma_j\}$$

II. Iterative Token-level Prompt Compression

Algorithm 2 Pseudo code of Iterative Token-level Prompt Compression (ITPC).

Input: A small language model $\mathcal{M}_s$; the prompt from budget controller $\mathbf{x}' = (\mathbf{x}^{\text{ins}}, \mathbf{x}^{\mathcal{D}}, \mathbf{x}^{\text{que}})$; target compression rate τ , adjusted compression rate $\Delta\tau_{\text{ins, que}}$.

- 1: Set the selected token set $\mathcal{T} = \phi$
- 2: Get segment set $\mathcal{S}$.
- 3: **for** $i = 1, 2, \dots, m$ **do**
- 4: Get the conditional probabilities $p(s_i)$ via Eq.(5)
- 5: Get the compression threshold γ_i with Eq.(6)
- 6: Append the compressed token to $\mathcal{T}$ via Eq.(7)
- 7: **end for**
- 8: Concatenate all tokens in $\mathcal{T}$ as $\tilde{\mathbf{x}}$.

Output: The compressed prompt $\tilde{\mathbf{x}}$.

$$p(\tilde{s}_j) = \prod_{i=1}^{\sum_k^j \tilde{L}_{s,k}} p(\tilde{s}_{j,i} | \tilde{s}_{j,<i}, \tilde{s}_{<j})$$

$$\approx \prod_{i=1}^{L_{s,j} + \sum_k^{j-1} \tilde{L}_{s,k}} p(s_{j,i} | s_{j,<i}, \tilde{s}_{<j}),$$

$$\tau_{s_j} = \begin{cases} \tau_{\text{ins}} + \Delta\tau, & \text{if } s_j \text{ from } \mathbf{x}^{\text{ins}}, \\ \tau_{\text{dems}}, & \text{if } s_j \text{ from } \mathbf{x}^{\mathcal{D}}, \\ \tau_{\text{que}} + \Delta\tau, & \text{if } s_j \text{ from } \mathbf{x}^{\text{que}}. \end{cases}$$

$$\tilde{s}_j = \{s_{j,i} | p(s_{j,i}) > \gamma_j\}$$

Experimental Settings

- **Data**

ShareGPT

ShareGPT is a dataset which users share conversations with ChatGPT indifferent languages and invarious scenario (i.e. chitchat, assistant and code)

yn2eWCt

[["human", "how do I add multiple new columns in m for power query or power bi?"], ["gpt", "In Power Query or Power BI, you can add multiple new columns by using the \"Add Column\" menu in the Home tab. Once you click on the \"Add Column\" menu, you will ...

Arxiv -March23

A dataset consisting of latest academic papers created in March 2023 from the arXiv preprint repository.

<http://arxiv.org/abs/2303.06806v1>

202303130
12855

Neural Diarization
with Non-
autoregressive
Intermediate
Attractors

["Yusuke Fujita",
"Tatsuya Komatsu",
"Robin Scheibler",
"Yusuke Kida",
"Tetsuji Ogawa"]

eess.AS

["eess.A
S",
"cs.CL",
"cs.SD"]

Boosting Source Code Learning with Data
Augmentation: An Empirical Study Jianjun Zhao =
===== End-to-end neural diarization (EEND) with
encoder-decoder-based attractors (EDA) is a
promising method to handle the whole speaker
diarization problem simultaneously with a single
neural network. While the EEND model can
produce all fram ...

Experimental Settings

- **Data**

GSM8K

GSM8K is a dataset of 8.5K high quality linguistically diverse grade school math word problems created by human problem writers. The dataset is segmented into 7.5K training problems and 1K test problems.

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

Natalia sold $48/2 = \ll 48/2=24 \gg 24$ clips in May. Natalia sold $48+24 = \ll 48+24=72 \gg 72$ clips altogether in April and May. ##### 72

BBH

BIG Benchmark (Beyond the Imitation Game Benchmark) In traditional NLP, 200 diverse text-based tasks such as mathematics, commonsense reasoning, and question-answering. BIG-Bench Hard (BBH) is a subset of 23 particularly **challenging BIG-Bench tasks**.
- Multiple-choice qa, open-domain qa, multi-label classification etc.

Input

Target

True and not not (not False) is

True

Experimental Settings

- **Baselines**
- **GPT-4 Generation**
 - instruct the GPT-4 to compress the original prompt.
- **Random Selection**
 - randomly selecting the demonstrations or sentences of the original prompt
- **Selective-Context (Li, 2023)**
 - phrase-level self information to filter out less informative content.
 - same small LM (i.e. Alpaca-7B)

Main Results

compression ratio

Methods	ShareGPT							Arxiv-March23						
	BLEU	Rouge1	Rouge2	RougeL	BS F1	Tokens	$1/\tau$	BLEU	Rouge1	Rouge2	RougeL	BS F1	Tokens	$1/\tau$
Constraint I	<i>2x constraint</i>							<i>350 tokens constraint</i>						
Sentence Selection	28.59	46.11	31.07	37.94	88.64	388	1.5x	22.77	50.1	25.93	33.63	88.21	379	4x
Selective-Context	25.42	46.47	29.09	36.99	88.92	307	1.9x	21.41	51.3	27.94	36.73	89.60	356	4x
Ours	27.36	48.87	30.32	38.55	89.52	304	1.9x	23.15	54.21	32.66	42.74	90.33	345	4x
Constraint II	<i>3x constraint</i>							<i>175 tokens constraint</i>						
Sentence Selection	18.94	35.17	18.96	26.75	85.63	255	2.3x	12.41	38.91	14.25	26.72	87.09	229	7x
Selective-Context	15.79	38.42	20.55	28.89	87.12	180	3.3x	12.23	42.47	19.48	29.47	88.16	185	8x
Ours	19.55	40.81	22.68	30.98	87.70	177	3.3x	13.45	44.36	24.86	34.94	89.03	176	9x

(+2.41) (+2.13)

(+5.38) (+5.47)

9x

Main Results

Methods	GSM8K			BBH		
	EM	Tokens	1/ τ	EM	Tokens	1/ τ
Full-shot	78.85	2,366	-	70.07	774	-
<i>1-shot constraint</i>						
1-shot	77.10	422	6x	69.60	284	3x
Selective-Context	53.98	452	5x	54.27	276	3x
GPT4 Generation	71.87	496	5x	27.13	260	3x
Ours	79.08	446	5x	70.11	288	3x
<i>half-shot constraint</i>						
Sentence Selection	72.33	230	10x	39.56	175	4x
Selective-Context	52.99	218	11x	54.02	155	5x
GPT4 Generation	68.61	223	11x	27.09	161	5x
Ours	77.41	171	14x	61.60	171	5x
<i>quarter-shot constraint</i>						
Sentence Selection	66.67	195	12x	46.00	109	7x
Selective-Context	44.20	157	15x	47.37	108	7x
GPT4 Generation	56.33	188	20x	26.81	101	8x
Ours	77.33	117	20x	56.85	110	7x
zero-shot	48.75 [†]	11	215x	32.32	16	48x
Simple Prompt	74.9	691	3x	-	-	-

(-1.44)

(-1.52)

Main Results

Methods	GSM8K			BBH		
	EM	Tokens	1/ τ	EM	Tokens	1/ τ
Full-shot	78.85	2,366	-	70.07	774	-
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BBH
challenging reasoning
tasks -> drop

Analysis

Compressed Prompt:

Follow the given examples and answer the question.

: Sam bought a dozen boxes each 30 high pens inside, \$10 each. He reanged five of boxes into of six each \$3 per. He sold the thelters separately at the of three \$2. much make total,

Lets think step

bought boxes x0 oflters

He 2 3ters in

Sam then boxes 6lters/box 0ters

He sold these boxes 5

Afterrelling these boxes there 36030lters

ese00 of three

sold groups2 each so made *2 \$20 from

In total, he015

Since his he \$ - \$120 = \$115 in profit.

The answer is 115

Q: Toula went to the bakery and bought various types of pastries. She bought 3 dozen donuts which cost \$68 per dozen, 2 dozen mini cupcakes which cost \$80 per dozen, and 6 dozen mini cheesecakes for \$55 per dozen. How much was the total cost?

Reference Answer(4-steps):

The total charge for the doughnuts was $3 \times \$68 = \$\langle 3*68=204 \rangle 204$.

The total charge for the mini cupcakes was $2 \times \$80 = \$\langle 2*80=160 \rangle 160$.

The total charge for the mini cheesecakes was $6 \times \$55 = \$\langle 6*55=330 \rangle 330$.

Therefore the total amount Toula paid for the pastries was $\$204 + \$160 + \$330 = \$\langle 204+160+330=694 \rangle 694$.

The Generation of Compressed Prompt(4-steps):

Toula bought 3 dozen donuts which cost \$68 per dozen. So, she spent $3 \times 68 = \$204$ on donuts.

She also bought 2 dozen mini cupcakes which cost \$80 per dozen. So, she spent $2 \times 80 = \$160$ on mini cupcakes.

Finally, she bought 6 dozen mini cheesecakes for \$55 per dozen. So, she spent $6 \times 55 = \$330$ on mini cheesecakes.

Therefore, the total cost was $204 + 160 + 330 = \$694$.

The answer is $\boxed{694}$.

The Generation of Selective-Context:

A: Toula bought $3 \times 12 = \langle 3*12=36 \rangle 36$ donuts.

Then, the total cost of the donuts is $36 \times \$68 = \$\langle 36*68=2448 \rangle 2448$.

How many mini cupcakes did Toula buy in total?

She bought $2 \times 12 = \langle 2*12=24 \rangle 24$ mini cupcakes.

So, the total cost of the mini cupcakes is $24 \times \$80 = \$\langle 24*80=1920 \rangle 1920$.

How many mini cheesecakes did Toula buy in total?

She bought $6 \times 12 = \langle 6*12=72 \rangle 72$ mini cheesecakes.

Thus, the total cost of the mini cheesecakes is $72 \times \$55 = \$\langle 72*55=3960 \rangle 3960$.

Therefore, the total cost of all the pastries is $\$2448 + \$1920 + \$3960 = \$\langle 2448+1920+3960=8328 \rangle 8328$. Answer: $\boxed{8328}$.

- Especially on GSM8K, its performance is lower than ours by 33.10 points at a compression ratio of 20x.

Ablation

	EM	Tokens	$1/\tau$
Ours	79.08	439	5x
- w/o Iterative Token-level Prompt Compression	72.93	453	5x
- w/o Budget Controller	73.62	486	5x
- w/o Dynamic Compression Ratio	77.26	457	5x
- w/ Random Selection in Budget Controller	72.78	477	5x
- w/o Distribution Alignment	78.62	452	5x
- w/ Remove Stop Words	76.27	1,882	1.3x

Table 3: Ablation study on GSM8K in 1-shot constraint.

- They conjecture this variant may lose essential information in the prompt, especially for low-frequency keywords that frequently appear in the given prompt.
- Theirs w/ Random Selection in Budget Controller indicates that selecting sentences or demonstrations based on perplexity can better identify information-rich sentences for target LLMs.

Discussion

	EM	Tokens	$1/\tau$
Ours in 1-shot constraint	83.51	439	5x
Ours in half-shot constraint	82.61	171	14x
Simple Prompt	81.8	691	3x

	EM	Tokens	$1/\tau$
Ours with GPT2 in 1-shot constraint	77.02	447	5x
Ours with GPT2 in half-shot constraint	76.42	173	14x
Ours with GPT2 in quarter-shot constraint	76.27	128	18x

Conclusion

- They introduce LLMLingua, which is based on the small LM's PPL for black-box LLMs.
- They validate the effectiveness of their approach on 4 datasets from different domains, i.e., GSM8K, BBH, ShareGPT, and Arxiv-March23.
- Their approach holds substantial practical implications, as it not only reduces computational costs but also offers a potential solution for accommodating longer contexts in LLMs.

Demos

Instruction

Please reference the following examples to answer the math question,

Context

Question: Angelo and Melanie want to plan how many hours over the next week they should study together for their test next week. They have 2 chapters of their textbook to study and 4 worksheets to memorize. They figure out that they should dedicate 3 hours to each chapter of their textbook and 1.5 hours for each worksheet. If they plan to study no more than 4 hours each day, how many days should they plan to study total over the next week if they take a 10-minute break every hour, include 3 10-minute snack breaks each day, and 30 minutes for lunch each day?

Let's think step by step

Angelo and Melanie think they should dedicate 3 hours to each of the 2 chapters, $3 \text{ hours} \times 2 \text{ chapters} = 6 \text{ hours total}$.

For the worksheets they plan to dedicate 1.5 hours for each worksheet, $1.5 \text{ hours} \times 4 \text{ worksheets} = 6 \text{ hours total}$.

Angelo and Melanie need to start with planning 12 hours to study, at 4 hours a day, $12 / 4 = 3 \text{ days}$.

However, they need to include time for breaks and lunch. Every hour they want to include a 10-minute break, so $12 \text{ total hours} \times 10 \text{ minutes} = 120 \text{ extra minutes for breaks}$.

They also want to include 3 10-minute snack breaks, $3 \times 10 \text{ minutes} = 30 \text{ minutes}$.

And they want to include 30 minutes for lunch each day, so $120 \text{ minutes for breaks} + 30 \text{ minutes for snack breaks} + 30 \text{ minutes for lunch} = 180 \text{ minutes, or } 180 / 60 \text{ minutes per hour} = 3 \text{ extra hours}$.

So Angelo and Melanie want to plan $12 \text{ hours to study} + 3 \text{ hours of breaks} = 15 \text{ hours total}$.

They want to study no more than 4 hours each day, $15 \text{ hours} / 4 \text{ hours each day} = 3.75$

They will need to plan to study 4 days to allow for all the time they need.

The answer is 4

Question: Mark's basketball team scores 25 2 pointers, 8 3 pointers and 10 free throws. Their opponents score double the 2 pointers but half the 3 pointers and free throws. What's the total number of points scored by both teams added together?

Let's think step by step

Mark's team scores 25 2 pointers, meaning they scored $25 \times 2 = 50 \text{ points in 2 pointers}$.

Question

Question: Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Target Token (To use this, set Compression Ratio to 0)

200

Compression Ratio (To use this, set Target Token to -1)

0

Compress Prompt!

Demos

Compressed Prompts

compressed_prompt

Please reference the following examples to answer the math question,

Question: can buy 4 apples 1 for You bought 36 fruits evenly split between and of 1 \$ does cost if total bill \$
's think step

If were between 3 of then 1 363 = 12 of fruitIf 1 orange then oranges50 \$if 1 \$66 \$60 the 2 fruit
of is, and is then 1W=4if we know we bought 12 and 12W 12

that can the 12 =

of apple (60/The

: Sam a dozen with 30 highlighter pens each Heanged into3 the separately the. much
think boxes120 12 = 360Sam then took 5 boxes × 6 highlighters/box = 30 highlighters.
He sold these boxes for 5 * \$3 = \$15

After selling these 5 boxes there were 360 - 30 = 330 highlighters remaining.

These form 330 / 3 = 110 groups of three pens.

He sold each of these groups for \$2 each, so made 110 * 2 = \$220 from them.

In total, then, he earned \$220 + \$15 = \$235.

Since his original cost was \$120, he earned \$235 - \$120 = \$115 in profit.

The answer is 115

Question: Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs.
This increased the value of the house by 150%. How much profit did he make?

Saving

The tokens number of original prompt

2428

The tokens number of compressed prompt

331

Actual Compression Ratio

7.3x

Saving Cost

, Saving \$0.1 in GPT-4.

Implementation

Q & A

Thank You!